IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-379

1. Topic (12 words max)

EduMind: An Intelligent Cloud System for Academic Performance Analysis and Engagement Boosting

2. Research group the project belongs to

TIM - Technology Integration and Management

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

A major research problem in the current educational technology landscape is the limited capacity of existing LMS and AI-powered analytics systems to deliver real-time, personalized, and explainable learning experiences while maintaining data privacy and operational scalability. Although AI and machine learning models—such as deep learning and reinforcement learning—have shown strong predictive capabilities for student outcomes and engagement, their lack of transparency makes it difficult for educators and learners to trust and effectively use these predictions in practice[1][2][3]. This "black-box" nature of AI hinders actionable interventions and slows the adoption of advanced analytics in education.

Compounding this, educational data is often fragmented across multiple, siloed platforms, making comprehensive analysis and adaptive learning difficult. Privacy and security concerns are heightened as regulations such as GDPR demand strict data handling, and solutions like federated learning are still emerging in educational applications[4]. While cloud-based learning analytics platforms offer scalable, real-time adaptive capabilities, they also introduce new challenges related to cost, reliability, and accessibility, especially in under-resourced regions[5][6].

Furthermore, current engagement monitoring and intervention mechanisms in LMSs often lack the sophistication for timely, personalized support based on dynamic behavioral data, resulting in missed opportunities to prevent disengagement and optimize learning outcomes[7][8].

This research seeks to address these gaps by developing a cloud-native, AI-powered framework that integrates automated MLOps pipelines, explainable predictive analytics, adaptive learning style identification, and real-time engagement monitoring. The goal is to provide transparent, actionable insights and timely interventions for educators and learners, ultimately enhancing engagement, learning outcomes, privacy, and scalability1576823.

References:

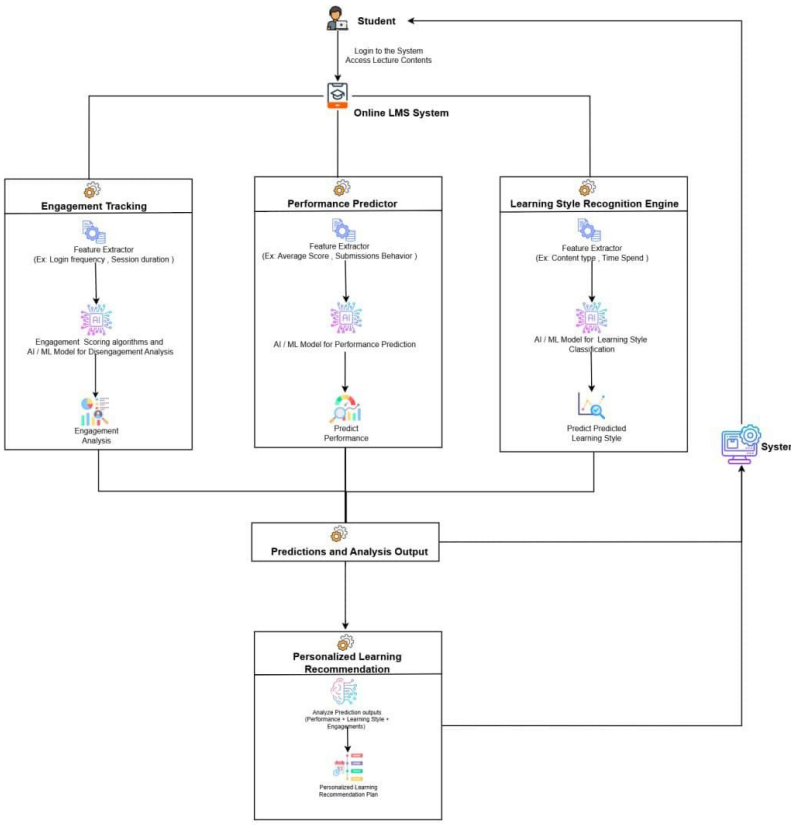
- 1 Khosravi, H. et al. (2022). Explainable Artificial Intelligence in education. ScienceDirect.
- 4 Mohammadi, S. et al. (2024). Balancing privacy and performance in federated learning: A systematic literature review on methods and metrics. Journal of Parallel and Distributed Computing.
- 5 BytePlus (2025). AI Cloud Computing for Education 2025.
- 7 Mendeley (n.d.). Reinforcement Learning in Education: A Literature Review.
- 6 Savvycom (2025). What's In 2025: Learning Analytics In A Nutshell.
- 8 EPALE (2025). Crafting personalized learning paths with AI for lifelong learning: a systematic literature review.
- 2 Türkmen, G. (2025). The Review of Studies on Explainable Artificial Intelligence in Education. SAGE Journals.
- 3 Altukhi, Z.M. (2024). Explainable AI Definitions and Challenges in Education. University of Technology Sydney.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a **cloud-native, AI-powered intelligent learning analytics platform** designed to enhance learner engagement, performance, and personalization in a scalable, cross-platform environment. The platform integrates real-time data processing, machine learning, reinforcement learning, explainable AI, and cloud DevOps practices to deliver a seamless and adaptive learning experience.

It consists of four interlinked components: (1) a **Proactive Engagement Optimization System** that predicts student disengagement using real-time activity data and delivers timely, personalized interventions; (2) a **Learning Style Recognition Engine** that continuously analyzes learner behavior to identify and adapt content to individual cognitive preferences; (3) an **Explainable AI Prediction Suite** that forecasts academic performance and visualizes the reasoning behind predictions to improve trust and decision-making; and (4) a robust **AI-Powered MLOps Pipeline** that automates model training, deployment, monitoring, and retraining based on performance or drift.

for a single institution, but for **diverse and decentralized learners** across MOOCs, informal learning tools, and LMSs — making it uniquely positioned to support modern, open-access education.



7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

This research requires a combination of expertise in **artificial intelligence (AI)**, **machine learning (ML)**, **cloud computing**, and **education technology**. Knowledge of **learning management systems (LMS)** and curriculum structuring supports effective data collection, interpretation, and content adaptation aligned with pedagogical goals.

Expertise in **cloud-based infrastructures** is necessary for the scalable deployment of the platform, ensuring that it can handle real-time data processing and predictions on a global scale. Familiarity with cloud services such as **AWS**, **Azure**, and **Google Cloud** is required to manage data pipelines, machine learning models, and scalable storage solutions.

Expertise in developing predictive models, reinforcement learning algorithms, and explainable AI techniques (e.g., SHAP, LIME) is essential to build adaptive systems that continuously learn from student interactions and provide transparent, trustworthy recommendations.

This multidisciplinary expertise and comprehensive data infrastructure are essential for building a functional and scalable AI-powered learning analytics platform.

Data requirements for this project include access to various educational datasets that provide insights into student behavior, performance, and engagement. This includes data from **learning management systems (LMS)**, **student performance records**, **attendance systems**, and **student interaction logs**. These data will be processed and analyzed to identify patterns, predict future performance, and personalize learning interventions. **Ethical considerations** regarding data privacy and security are paramount, especially for sensitive student information, necessitating adherence to privacy regulations like **GDPR** or **FERPA**.

Additionally, the project will require labeled datasets to train machine learning models, particularly for **classification tasks** (e.g., predicting dropout risk or academic performance). Data on student performance, quiz scores, assignment submission times, and interaction frequency will be crucial for training the **reinforcement learning models** for real-time personalization.

8. Objectives and Novelty

Main Objective The main objective of this project is to develop an AI-powered, cloud-based learning analytics platform that enables real-time student performance prediction, adaptive content personalization, and proactive engagement interventions to improve learning outcomes and educational efficiency.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Basnayaka B M V C IT22278944	Implementing a fully automated, cloud-native MLOps pipeline to manage the model lifecycle (training, deployment, drift monitoring, and self-healing retraining) is a significant undertaking that requires a combination of architectural design, specific tool selection, and	1. Define Your Goals & Requirements: - What kind of models will you deploy? (e.g., real-time, batch, small, large) - What is your performance (latency, throughput) and scalability needs? - What are the compliance and security requirements? - What existing cloud infrastructure are you using? 2. Architectural Design: - Sketch out the flow of data and models through your pipeline. - Decide on your cloud provider (AWS, GCP, Azure) and	This research introduces an intelligent AI-powered chatbot designed to detect, analyze, and prioritize errors within an MLOps pipeline deployed on MCP (Managed Cloud Platform) servers. Unlike conventional monitoring systems that merely display logs and alerts, the proposed chatbot leverages natural language understanding (NLU) and log analysis to provide **context-aware error detection**, **root cause identification**, and **proactive recommendations** for remediation. The system integrates with the **LGTM (Loki, Grafana, Tempo, Mimir) stack**, where both application and metric

	robust engineering practices.	the specific services you'll leverage. - Choose between managed ML services or a more open-source, Kubernetes-centric approach (often a hybrid is best). 3. Set Up Version Control & Data Management: - Establish a Git repository structure for your ML projects. - Implement data versioning (e.g., DVC, or use cloud storage with versioning and careful naming conventions). - Consider setting up a Feature Store early for larger projects. 4. Develop Your Initial ML Model & Training Script: - Write modular and reproducible training code. - Containerize your training environment (Dockerfile). 5. Build the CI Pipeline (Training Automation): - Trigger: Push code changes to the training repository. - Steps:	logs from the MLOps pipeline are collected and stored. By tapping into Grafana MCP's centralized observability features, the chatbot parses these logs in real-time, prioritizes incidents based on severity and system impact, and generates intelligent alerts and recommendations directly to the operations team via chat interfaces (e.g., Slack, MS Teams). Through this novel integration, the solution aims to significantly **reduce mean time to detect (MTTD)** and **mean time to resolve (MTTR)** MLOps pipeline issues, fostering a more autonomous, resilient, and developer-friendly operational ecosystem.

		- Linting and unit tests for training code. - Build the Docker image for the training environment. - Execute the training job using the containerized code on scalable infrastructure (e.g., Kubernetes, SageMaker Training Jobs). - Log experiment details (metrics, hyperparameters, artifacts) to the experiment tracker (e.g., MLflow). - Register the trained model in the model registry if it meets initial quality gates. - Tools: Use your chosen CI/CD orchestrator (e.g., GitHub Actions, GitLab CI, Kubeflow Pipelines). 6. Build the CD Pipeline (Deployment Automation): - Trigger: New model version registered in the model registry, or manual approval. - Steps: - Run integration tests and end-to-end tests against the new model.	

		- Perform A/B testing or canary deployments in a staging environment. - If successful, promote the model to production. - Update the serving endpoint to use the new model version. - Automate rollback mechanisms in case of issues. - Tools: CI/CD orchestrator, cloud-native deployment services (e.g., Kubernetes Deployments, SageMaker Endpoints). 7. Implement Model Monitoring: - Data Collection: Capture inference requests and responses (input data, predictions, actual outcomes if available). - Metrics: Monitor model performance metrics (accuracy, precision, recall, F1-score), data drift (e.g., population stability index, KS statistic), concept drift, latency, throughput, resource utilization.	

		- Alerting: Set up alerts for significant deviations in these metrics. - Tools: Specialized ML monitoring tools, cloud monitoring services. 8. Enable Self-Healing Retraining: - Feedback Loop: Integrate alerts from the monitoring system with your CT pipeline. - Trigger: An alert (e.g., significant data drift detected, model performance drops below a threshold) triggers the CT pipeline. - CT Pipeline Steps: - Fetch new data (often from a real-time feedback stream). - Retrain the model using the latest data. Evaluate the retrained model against current production data and a new validation set. Compare the retrained model's performance against the current production model. If the retrained model is significantly better and passes	

		all quality gates, automatically deploy it to production (using the CD pipeline). - Log the retraining event and new model version.	
Bandara K A R U IT22354174	Implement Explainable AI-Driven Prediction Suite	• Develop supervised machine learning models to predict student outcomes such as **dropout risk** and **academic performance** using data from LMS, quizzes, assignments, and engagement records. • Collect and preprocess relevant student data, including academic performance indicators, behavioral logs, attendance, and participation metrics. • Integrate **explainable AI techniques** such as **SHAP** and **LIME** to interpret model predictions both at the global level (overall feature importance) and at the local level (individual prediction explanations). • Generate SHAP summary plots and LIME visualizations to highlight which features most influence each prediction, enhancing the **transparency and trust** of the system.	Provides interpretable predictions using SHAP/LIME, making AI decisions transparent for educators. Enables proactive support through explainable visual analytics. Integrates explainability in real-time for trust and ethical AI adoption in education.

		• Design and implement an **interactive dashboard** using tools like **Dash** (Python) or **Power BI**, which will visualize: • Predicted outcomes (e.g., at-risk status or performance level) • Key contributing features and their impact • Filter and search functionality for specific students or classes • Enable **real-time or near-real-time insights** by connecting the system to live student data feeds or LMS APIs, ensuring that predictions and visualizations are constantly updated. • Empower educators to take **proactive, personalized interventions** based on the predictions and their explanations, such as targeted feedback, early warnings, or tailored support strategies	
Subasinghe S A V R IT22325846	Dynamically identify individual learning styles in real-time through behavioral analysis and adaptive	• Gather data from LMS, quizzes, videos, assignments, and participation to assess student behavior.	The Learning Style Recognition Engine is innovative because it uses real-time behavioral data to dynamically assess and adapt to each student's unique learning preferences, offering personalized

	machine learning to enable personalized content delivery.	• Research and define common learning styles (e.g., visual, auditory, kinesthetic) based on existing educational frameworks. • Develop features such as time spent on videos, responses to interactive content, and quiz performance to identify learning preferences. • Train a classification model (e.g., decision tree, neural network) to predict learning styles based on student behavior data. • Design the system to adjust learning content according to the identified learning style (e.g., visual aids for visual learners, audio materials for auditory learners). • ⍰ Evaluate the model's accuracy by comparing predicted learning styles with student feedback or	learning experiences. Unlike traditional systems that rely on self-reported learning styles, this component uses machine learning to objectively classify learning styles based on actual student behavior, ensuring more accurate and data-driven personalization.

		performance improvements.	
Gimhani K T K IT22317780	Prediction and analysis of student disengagement, with personalized intervention through automated recommendation of alternative learning	• Collection of student interaction data from the LMS including login frequency, session duration, content type usage, clickstream, forum participation, device type, and access time. • Preprocessing and cleaning of raw interaction data to handle missing values, normalize time series, and categorize content types. • Development of engagement scoring algorithms to quantify student activity levels using statistical and machine learning methods. • Design and implementation of predictive models to detect early signs of disengagement or struggle with specific learning concepts based on behavioral patterns and past engagement data. • Creation of visualization dashboards and heatmaps to display engagement scores and trends for educators and administrators. • Integration of personalized feedback mechanisms that	This system introduces a novel AI-driven engagement tracking platform that integrates real-time behavioral analytics with personalized learning interventions within an LMS environment. Unlike traditional engagement monitoring tools that only provide static reports, this platform predicts learner disengagement early by analyzing multifaceted interaction data such as login patterns, content interaction, and forum participation. The novelty lies in the system's ability to act as a self-evolving tutor by automatically recommending tailored alternative resources, micro-lessons, or practice problems when signs of struggle or disengagement are detected, without human intervention. This dynamic personalization enhances learner retention and outcomes by continuously adapting to individual learning needs. Additionally, the platform offers rich visual analytics, including heatmaps and engagement scoring dashboards, to

		automatically recommend alternative learning resources, micro-lessons, and practice problems when disengagement or difficulties are detected. • Evaluation of model accuracy and effectiveness through testing on LMS datasets and pilot student groups. • Deployment of the system as a platform with an intuitive interface to facilitate real-time monitoring and intervention without manual supervision.	support educators in monitoring student progress more effectively. The integration of predictive modeling with automated adaptive feedback represents a significant advancement over existing systems that lack proactive intervention capabilities. Developed a solution, platform facilitates integration with existing LMS infrastructure and provides real-time insights and support to both learners and educators, making it a unique and scalable tool in the field of intelligent learning analytics.

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Basnayaka B M V C IT22278944	This component leverages IT specialization by applying advanced automation, orchestration, and cloud infrastructure management. It ensures the seamless training, deployment, drift monitoring, and self-healing retraining of machine learning models using real-time feedback, which demonstrates expertise in scalable system design and operational reliability.
Bandara K A R U IT22354174	Forecasts student outcomes such as grades and dropout risk using machine learning models, while integrating dashboards that make predictions transparent and interpretable for both learners and educators. This supports informed decision-making and builds trust in the analytics process.
Subasinghe S A V R IT22325846	This feature is relevant to IT specialization by using machine learning and cloud technology to personalize learning based on a learner's preferred style (visual, auditory, or reading/writing). IT professionals often work with complex, evolving concepts, and this adaptive approach helps them better absorb and retain information. By dynamically adjusting content formats, it ensures more effective, engaging learning experiences, supporting skill development in areas like programming, networking, and system architecture.
Gimhani K T K IT22317780	This component applies core Information Technology concepts to design and develop a cloud-based intelligent engagement analytics system. It involves collecting and preprocessing large-scale user interaction data from the LMS, implementing predictive models using machine learning techniques, and generating personalized learning interventions. The system integrates backend data processing, real-time analytics, and frontend visualization through interactive dashboards. It uses full-stack development skills to connect the analytics engine with a user-friendly web interface. This component reflects key IT areas such as data analytics, machine learning, cloud computing, API integration, and system deployment to support scalable and intelligent education platforms.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	MS	Thamali	Dassanayake	26/05/2025

Co-Supervisor				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

This part is to be filled by the Topic Screening Staff members.

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	
Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.